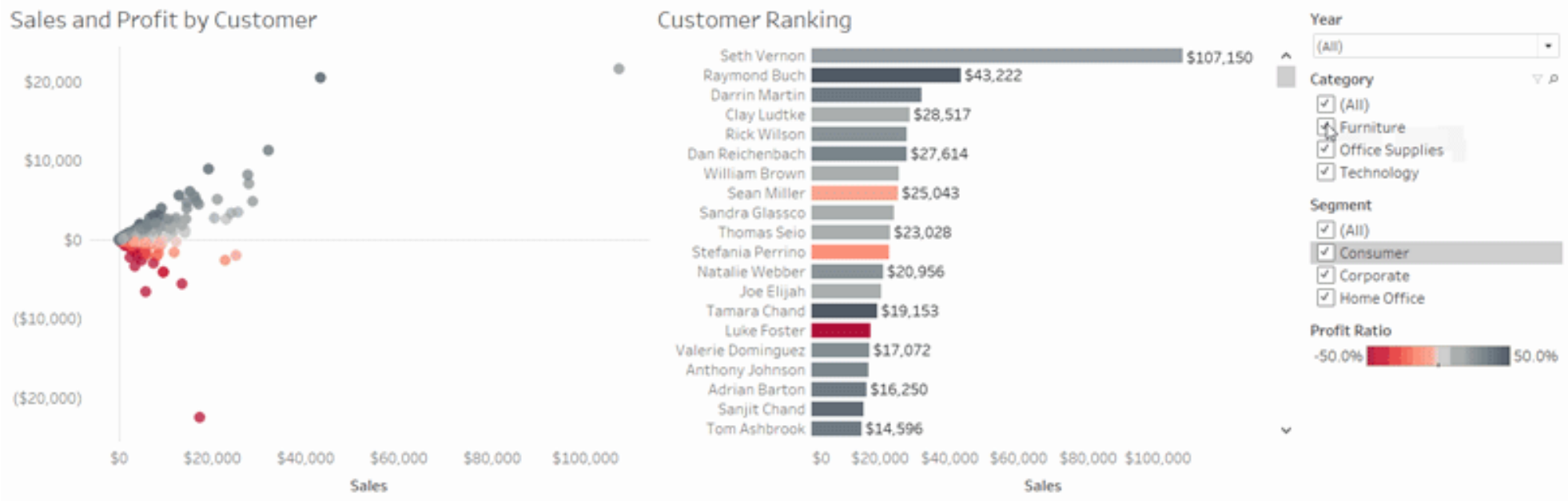


Intro to Tableau

Kellie Clinton (KeC)

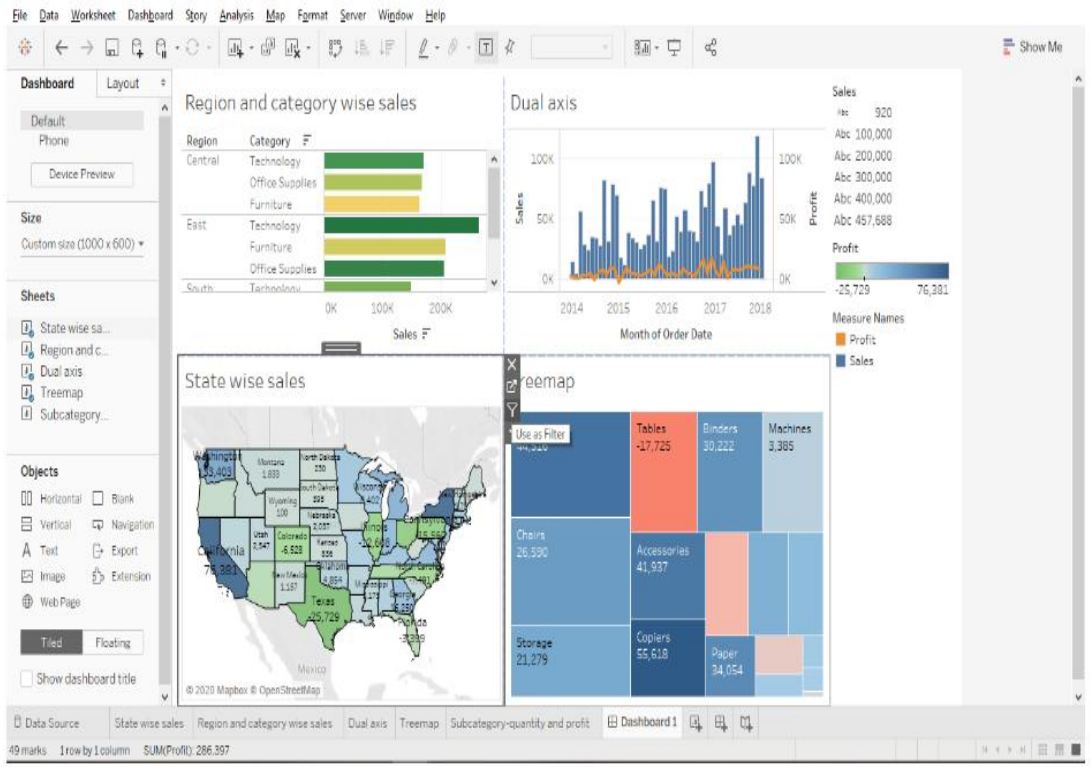
Agenda

- Overview
- Preparing Data
- Exploring Data
- Coding and Applications
- Key Takeaways

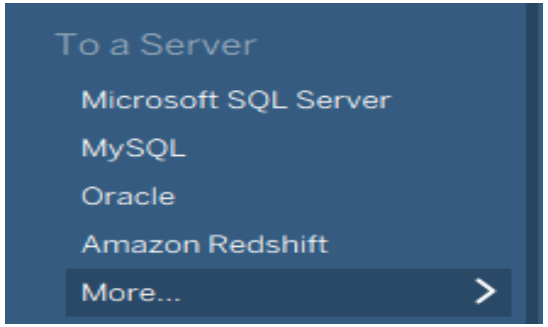


What is Tableau

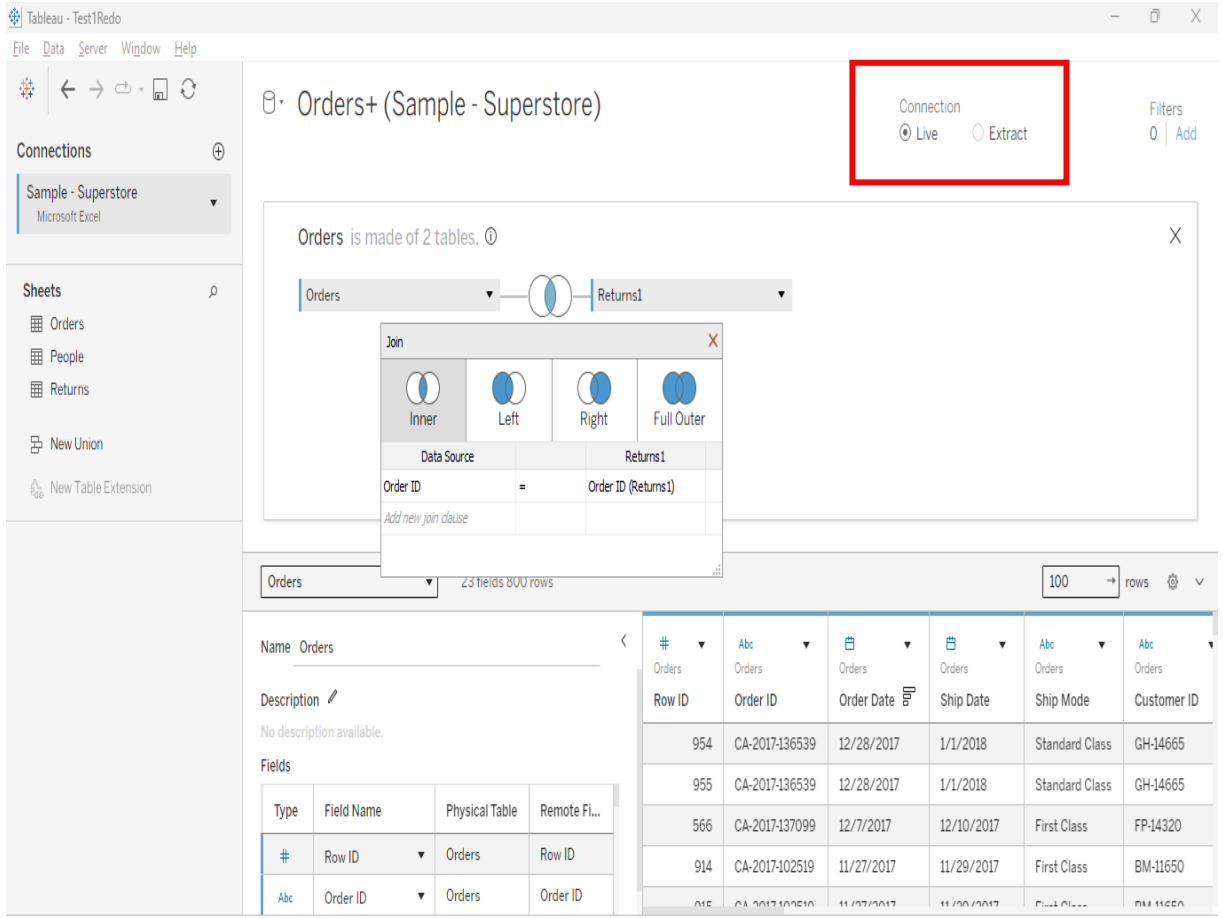
- Business intelligence and data visualization platform
- Creates interactive charts, graphs, and dashboards
- Shift the conversation from *"What does this data mean?"* to *"What should we do next?"*
- Team Collaboration
- Access: Enterprise Tableau license or free trial



Preparing Data: Superstore



- Trusted connections across enterprise data sources
- Built-in SQL database connectors including PostgreSQL, MySQL, Oracle, and Snowflake
- Streamline ETL by cleansing data anomalies and standardizing formatting
- Automate data refreshes



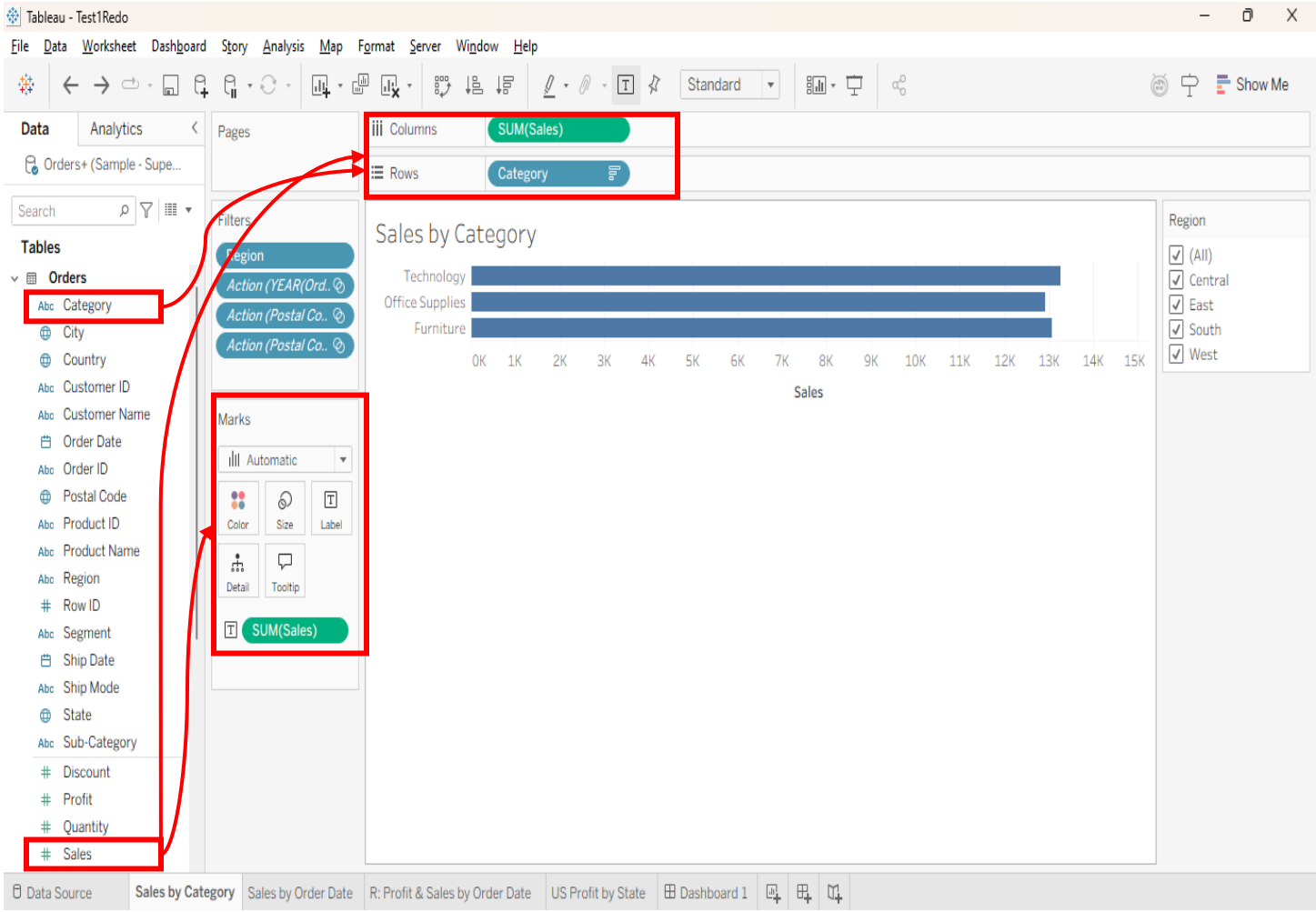
Exploring Data

- Dimensions describe categories or groupings
- Measures are quantitative values that can be analyzed
- Placing dimensions and measures into the worksheet determines how Tableau groups and summarizes the data



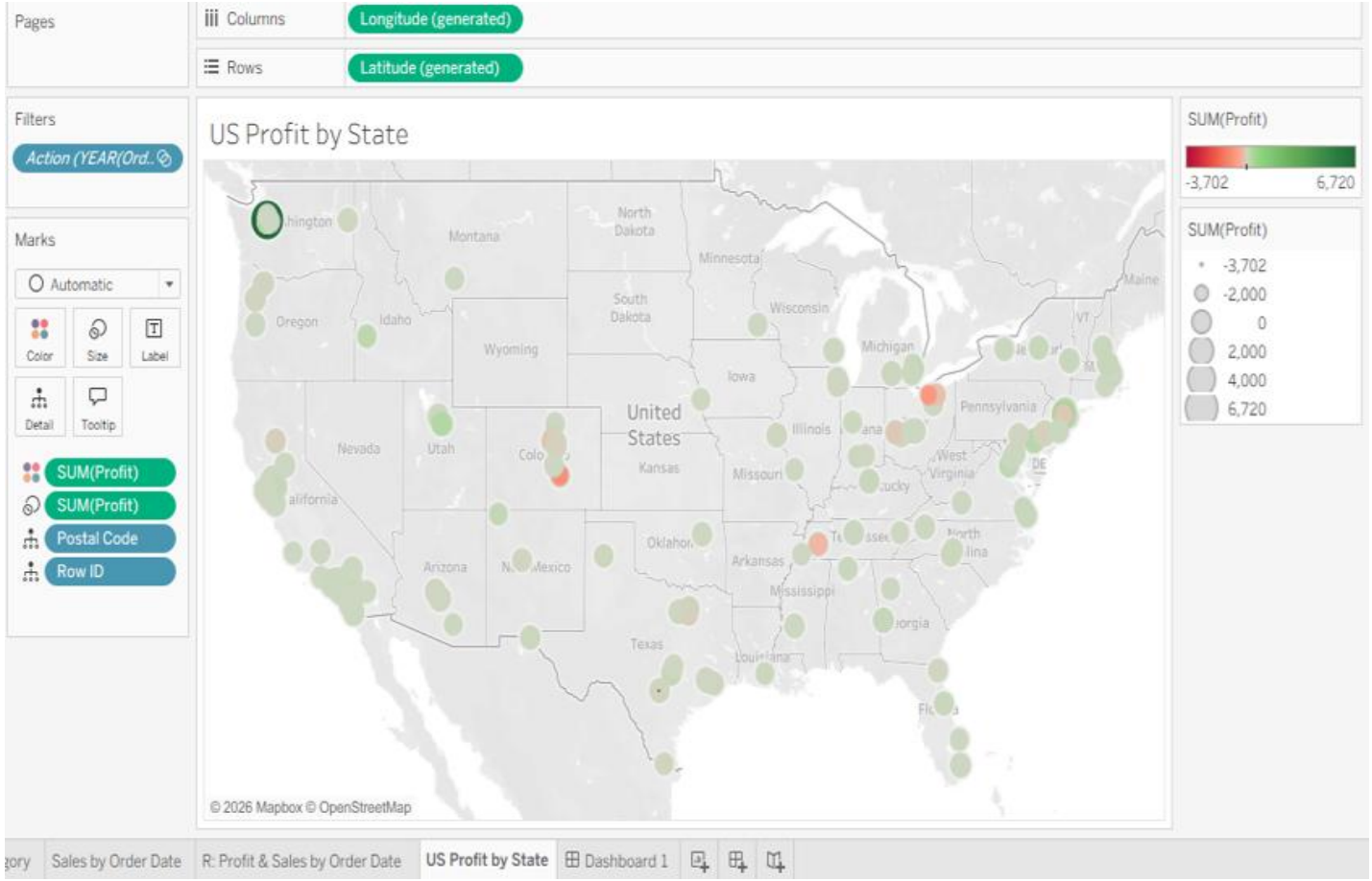
Visual: Sales by Category

- Which product category generates the most sales?
- Value: Helps the business understand where its sales are concentrated

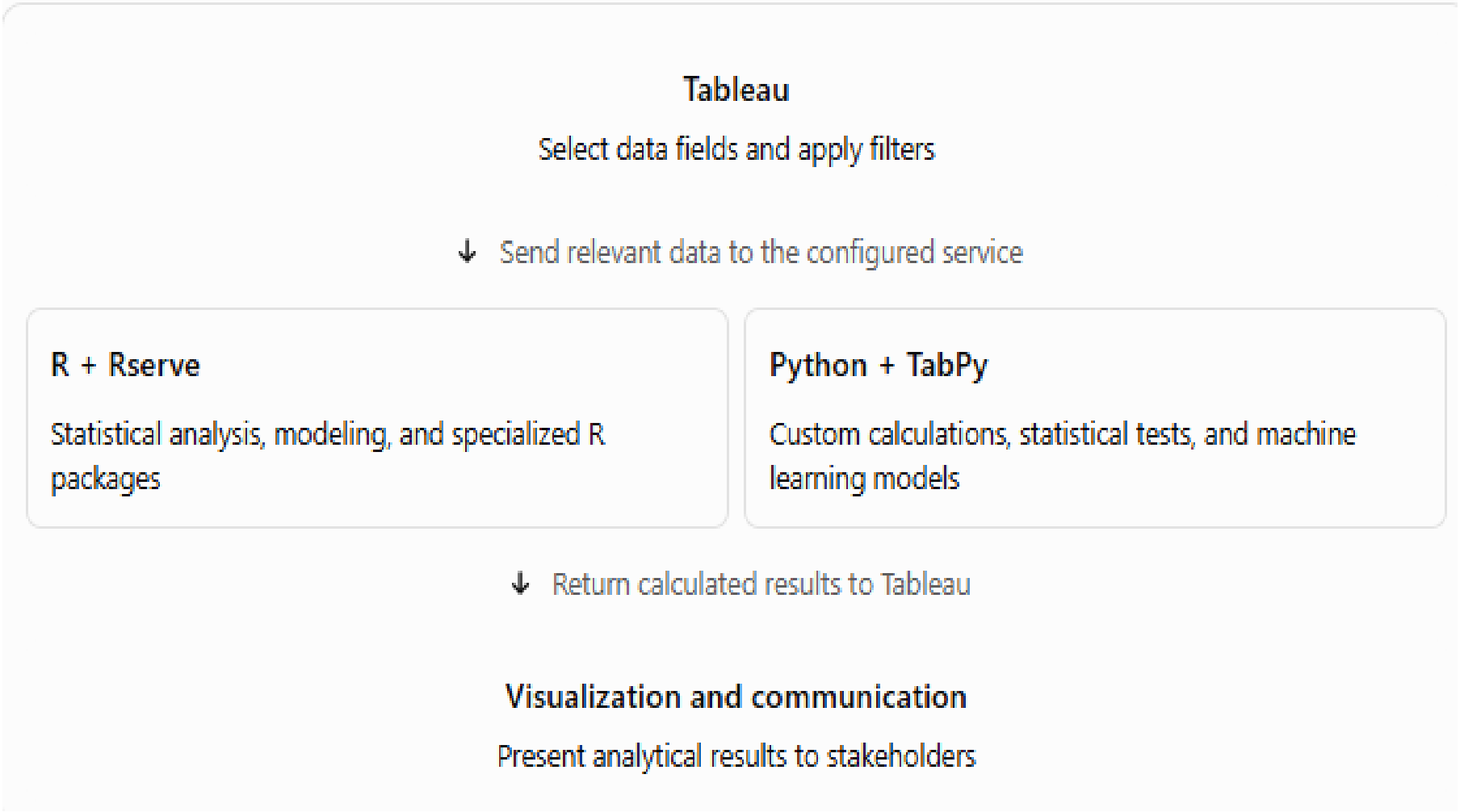


US Profit by Zip Code

- How does profitability vary across geographic areas?
- Value: Helps the business identify geographic differences in profitability and areas that may need further investigation

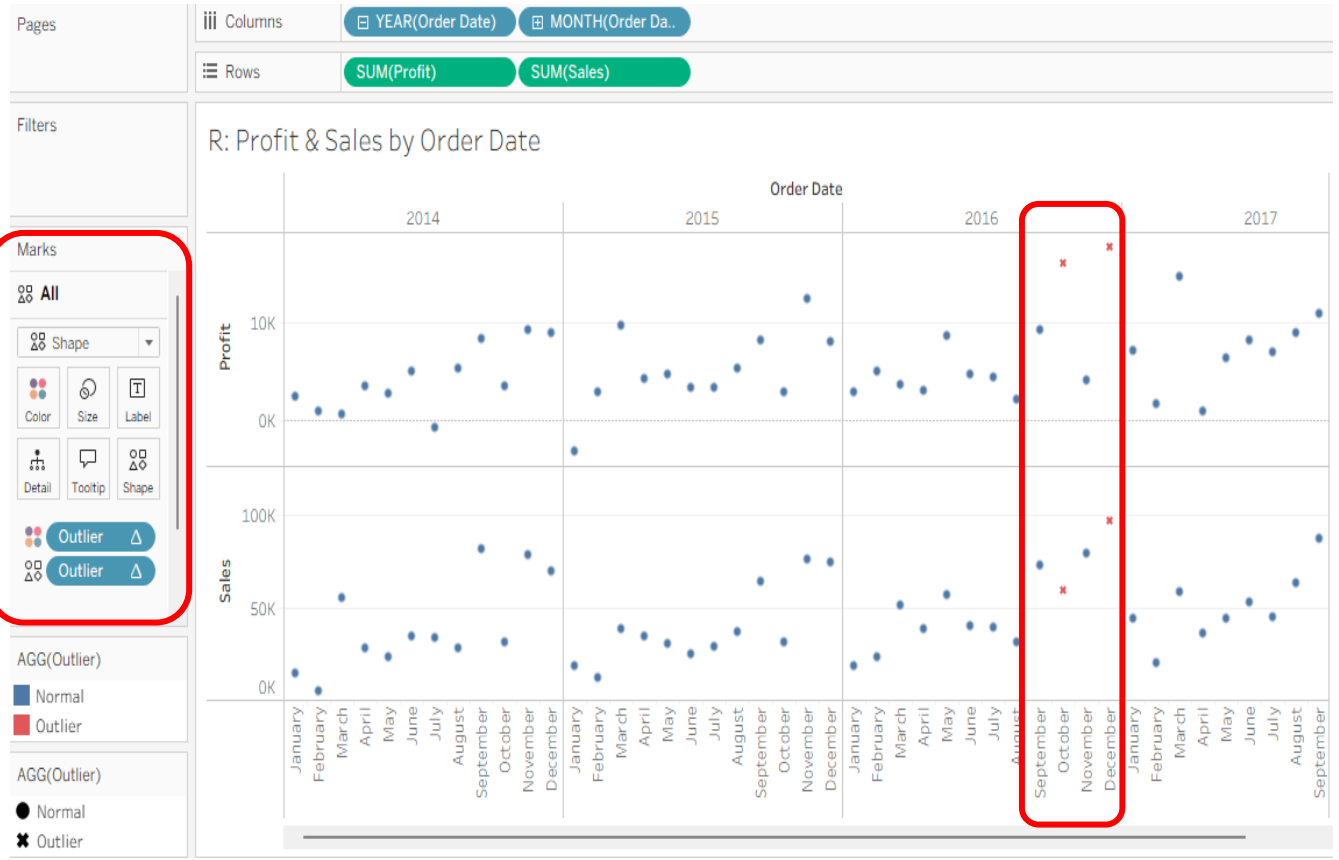


Coding and Applications: R & Python Integration



R: Profit & Sales by Order Date

- Does profit follow the same pattern as sales?
- Value: Helps identify periods that may need further investigation into discounts, costs, or product mix



Outlier

Results are computed along Table (across).

```
SCRIPT_REAL(
    "library(mvoutlier);
     # R Comment: Spatial sign algorithm returning 0 (outlier) or 1 (normal)
     sign2(cbind(.arg1))$wfinal01",

     // Tableau Comment: Data sent to R as '.arg1'
     sum([Profit])
) = 0
THEN "Outlier"
ELSE "Normal"
END
```

Python: Measuring the Relationship Between Sales and Profit

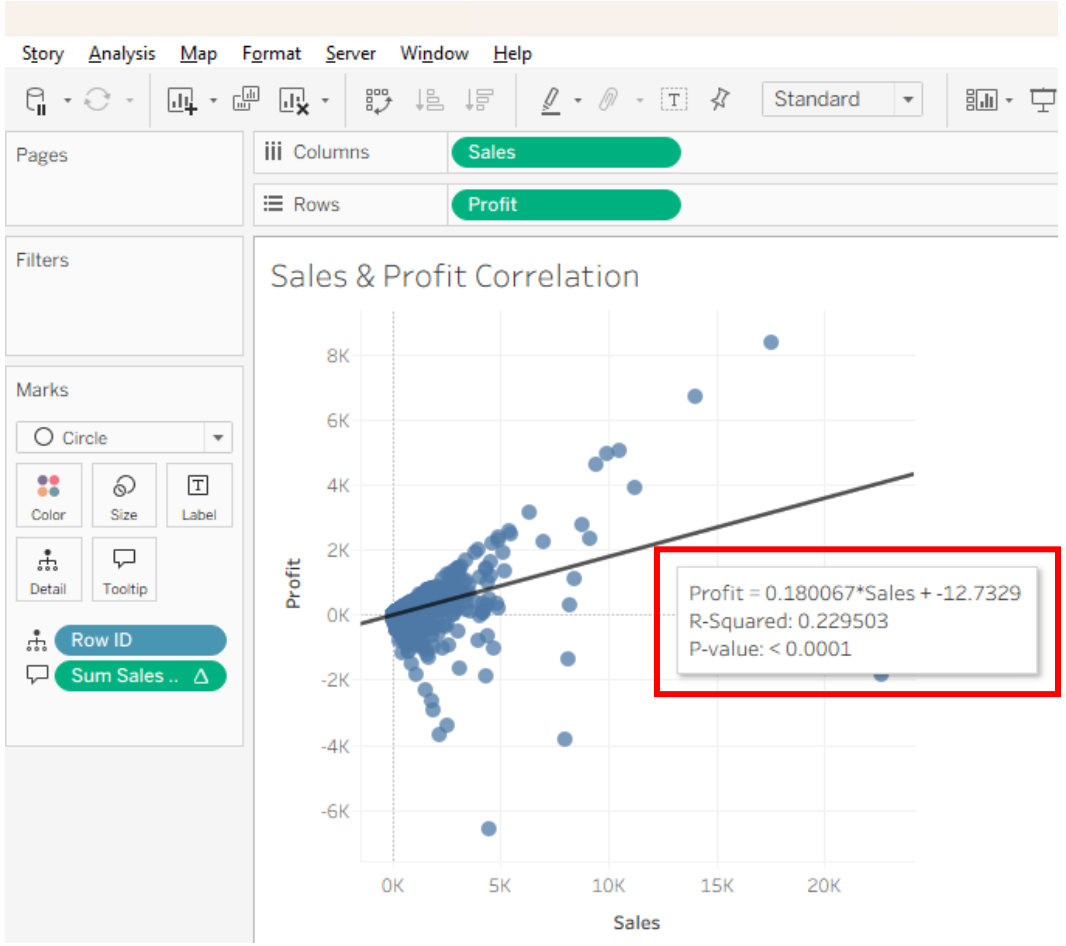
- Are sales and profit linearly associated?
- Value: Sales alone may not fully describe financial performance

Sum Sales & Profit

Results are computed along Table (across).

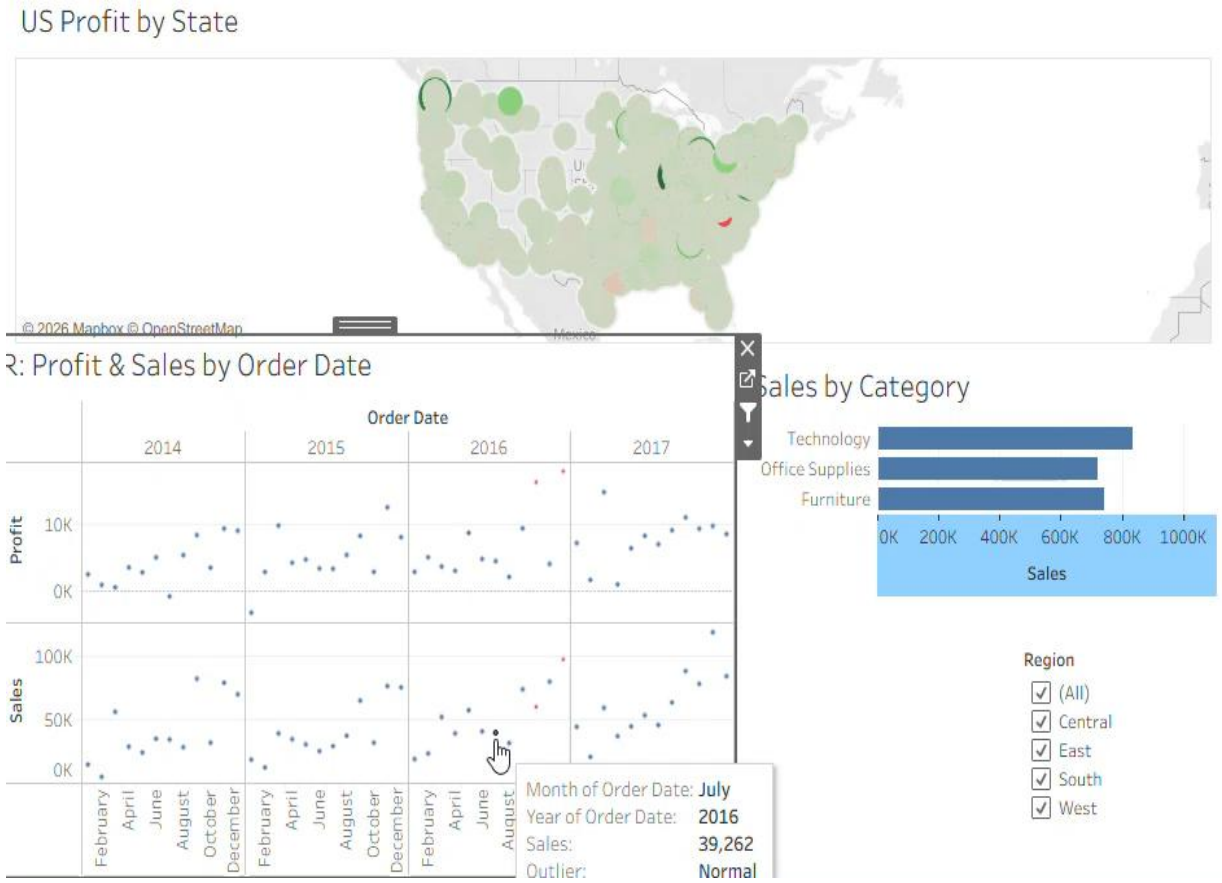
```
SCRIPT_REAL("
import numpy as np
# Calculate the single correlation value
corr = np.corrcoef(_arg1, _arg2)[0,1]

# Repeat that value for every single row so Tableau can map it
return [corr] * len(_arg1)
", SUM([Sales]), SUM([Profit]))
```



Bringing the Analysis Together: Dashboard

- Bringing multiple views together helps compare trends without switching between separate reports
- Stakeholders can use the dashboard to explore the available data themselves



When to Use Tableau

Strengths:

- Natively connects to enterprise databases, warehouses, and servers
- Rapid interactive data exploration
- Cross-team collaboration
- Impactful executive storytelling

Limitations:

- Cannot perform heavy machine learning, advanced NLP, or custom statistical architectures
- Inefficient for processing unstructured text (e.g., emails), running massive loop iterations
- Version-control and code-review workflow is more limited than a conventional Git-based workflow

Key Takeaways

- Self-service dashboards that non-technical clients and executive stakeholders can easily understand and trust
- Visual exploration, not heavy modeling
- Tool Integration combines SQL for data extraction, Python/R for advanced math, and Tableau for final visuals

Thank you!

Questions?